Coherence Sample Dataset

Exported from the Coherence AI learning app · October 22, 2026 · 2 lessons · 9 topic sections · 42 transcript chunks · 19 quiz questions

This document contains the sample learning data used by the app: lessons, their topic sections, timestamped transcript chunks, and the multiple-choice quiz bank. The same data is also available as CSV/JSON in coherence_sample_dataset.zip.

Note: the seeded lesson “Quantum mechanics as a framework” uses an approximate, unverified transcript written for demonstration — it is not the official MIT caption file. “Testing linear operators” is a small lesson imported through the app's own lesson-import feature during testing.

1. Lessons

Title	Source	YouTube ID	Duration
Quantum mechanics as a framework	MIT OpenCourseWare · 8.04 Quantum Physics I	jANZxzetPaQ	17:46
Testing linear operators	Test transcript supplied for verification	—	2:55

Quantum mechanics as a framework — A foundational MIT lecture introduces quantum mechanics and explains what it means for a theory to be linear.

Testing linear operators — Linear operators respect scaling and addition, which together give superposition. If L(u) = 0 and L(v) = 0, any linear combination of u and v is also a solution. Maxwell equations in vacuum are linear in the electric and magnetic fields, while nonlinear equations do not generally preserve superposition.

2. Topic sections

Lesson	Start	Title	Summary	Active
Quantum mechanics as a framework	00:00	Quantum mechanics as a framework	A framework for physics	True
Quantum mechanics as a framework	03:30	Defining linearity	What linearity means	True
Quantum mechanics as a framework	10:50	Scaling and adding solutions	Build a new solution	True
Testing linear operators	00:00	Scaling	A linear operator satisfies L(a u) = a L(u).	True
Testing linear operators	00:30	Addition	A linear operator satisfies L(u + v) = L(u) + L(v).	True
Testing linear operators	01:00	Superposition	Combining scaling and addition gives superposition.	True
Testing linear operators	01:30	Linear Combinations of Solutions	If L(u) and L(v) are zero, any linear combination of u and v is also a solution.	True
Testing linear operators	02:00	Maxwell Equations in Vacuum	Maxwell equations in vacuum are linear in electric and magnetic fields.	True
Testing linear operators	02:30	Nonlinear Equations	Nonlinear equations do not generally preserve superposition.	True

3. Transcript chunks

Quantum mechanics as a framework

00:00–00:30 Welcome to the course. Quantum mechanics is a framework for doing physics. It is not a single theory of one phenomenon; it is a set of general principles.
00:30–01:00 Quantum mechanics replaces classical physics as the fundamental description of nature. Classical mechanics is an approximation that works well for large objects.
01:00–01:30 The framework can be applied to many different physical phenomena: atoms, molecules, solids, nuclei, light and particle physics all use the same rules.
01:30–02:00 When you apply quantum mechanics to electromagnetism you get quantum electrodynamics. Applied to particles and forces you get quantum field theory.
02:00–02:30 So we should think of quantum mechanics as the language, and of particular theories as things written in that language for a specific system.
02:30–03:00 The principles of quantum mechanics have been tested to extraordinary precision and no experiment has contradicted them so far.
03:00–03:30 One of the key features of the framework is that it is a linear theory. Before we use that idea we need to define what linearity means.
03:30–04:00 Defining linearity. A theory is linear when its equations are linear in the unknown. We write the equation as an operator L acting on the unknown u equals zero.
04:00–04:30 The operator L is linear if L acting on a times u equals a times L acting on u, for any number a. Scaling the unknown scales the result.
04:30–05:00 And L acting on the sum u1 plus u2 equals L u1 plus L u2. The operator distributes over addition. These two properties together define a linear operator.
05:00–05:30 Maxwell's equations in vacuum are an example of a linear theory. The electric and magnetic fields appear only to the first power.
05:30–06:00 Because of this, the electromagnetic waves from two sources simply add. Two light beams can cross without disturbing each other.
06:00–06:30 An equation with a term like u squared, or u times its derivative, is nonlinear. Then the sum of two solutions is generally not a solution.
06:30–07:00 Classical mechanics often leads to nonlinear equations. Newton's laws with realistic forces are typically nonlinear, which makes them hard to solve.
07:00–07:30 Let us look at a simple example: the derivative of u with respect to time plus one over tau times u equals zero. Here L is d over dt plus one over tau.
07:30–08:00 Check the two properties: the derivative of a times u is a times the derivative of u, and the derivative of a sum is the sum of the derivatives. So L is linear.
08:00–08:30 Now add a term u squared to the equation. L acting on a times u produces a squared times u squared, which is not a times L u. Linearity fails.
08:30–09:00 In quantum mechanics the central equation is the Schrodinger equation, and it is linear in the wavefunction. This is a deep and exact statement.
09:00–09:30 Physicists have looked for small nonlinear corrections to the Schrodinger equation and have found none. Quantum mechanics appears to be exactly linear.
09:30–10:00 Linearity is what lets us talk about superposition. It is the mathematical reason quantum states can be combined.
10:00–10:30 To summarize the definition: a linear operator respects scaling and respects addition. Keep both conditions in mind.
10:30–11:00 Now we will see the most important consequence of linearity, which is how we build new solutions from old ones.
10:50–11:20 Scaling and adding solutions. Suppose u1 is a solution, so L u1 equals zero. Then a times u1 is also a solution, because L of a u1 is a times zero, which is zero.
11:20–11:50 Suppose u2 is another solution. Then u1 plus u2 is a solution too, since L of the sum is L u1 plus L u2, which is zero plus zero.
11:50–12:20 Combining both, any linear combination a1 u1 plus a2 u2 is a solution. This is the superposition principle for linear equations.
12:20–12:50 This lets you build a new solution from known ones. If you find a few simple solutions you can form infinitely many more.

12:50–13:20 For nonlinear equations this fails. The sum of two solutions is not a solution, so you cannot assemble complicated solutions from simple pieces.
13:20–13:50 In quantum mechanics, if two wavefunctions describe possible states of a system, then their sum, with complex coefficients, is also a possible state.
13:50–14:20 The coefficients can be complex numbers. Complex numbers are not optional in quantum mechanics; they are built into the framework.
14:20–14:50 This superposition of states is very different from the superposition of waves in classical physics. It concerns the state of a single system.
14:50–15:20 An electron can be in a superposition of being here and being there. A measurement finds it in one place, with probabilities set by the coefficients.
15:20–15:50 We will return to the meaning of these superpositions many times in the course. For now the key message is the mathematical one.
15:50–16:20 Linearity allows scaling and adding solutions, and that is why superposition is possible in quantum mechanics.
16:20–16:50 In the next lectures we will look at experiments, such as the photon and polarizers, that show why superposition is needed.
16:50–17:20 Let me review: quantum mechanics is a framework, the theory is linear, and linear combinations of solutions are solutions.
17:20–17:50 That completes the introduction. Next time we start with the photoelectric effect and the particle nature of light.

Testing linear operators

00:00–00:30 A linear operator respects scaling: L(a u) equals a L(u).
00:30–01:00 A linear operator respects addition: L(u plus v) equals L(u) plus L(v).
01:00–01:30 Combining scaling and addition gives superposition.
01:30–02:00 If L(u) and L(v) equal zero, any linear combination is also a solution.
02:00–02:30 Maxwell equations in vacuum are linear in electric and magnetic fields.
02:30–02:55 Nonlinear equations do not generally preserve superposition.

4. Quiz questions

Quantum mechanics as a framework

Q1. [Defining linearity · 03:30] According to the transcript, when is a theory linear?

A. When its equations contain no unknowns.
B. When its equations are linear in the unknown.
C. When its equations apply only to electric fields.
D. When its equations cannot be written using an operator.
Correct answer: B
A theory is linear when its equations are linear in the unknown. The transcript writes the equation as L acting on u equals zero.

Q2. [Defining linearity · 04:00] Which pair of properties defines a linear operator L?

A. L(au)=aL(u) and L(u1+u2)=L(u1)+L(u2)
B. L(u)=u and L(0)=1
C. $L(u^2)=L(u)^2$ only
D. L is always a derivative
Correct answer: A
Linearity means respecting scaling and addition.

Q3. [Defining linearity · 04:00] Which equation expresses the scaling property of a linear operator L for any number a?

A. L(au) = L(u)
B. L(au) = a + L(u)
C. $L(au) = a^2L(u)$
D. L(au) = aL(u)
Correct answer: D
Scaling the unknown by a scales the result by the same number: L(au) = aL(u).

Q4. [Defining linearity · 04:30] Along with the scaling property, which property defines a linear operator?

A. L(u1 + u2) = L(u1) + L(u2)
B. L(u1 + u2) = L(u1)L(u2)
C. L(u1 + u2) = L(u1)
D. L(u1 + u2) = 0 for any u1 and u2
Correct answer: A
A linear operator distributes over addition. This addition property and the scaling property together define a linear operator.

Q5. [Defining linearity · 08:00] Which equation is NOT linear?

A. $du/dt + u/\tau = 0$
B. $du/dt + u/\tau + u^2 = 0$
C. Maxwell's equations in vacuum
D. The Schrödinger equation
Correct answer: B
The u^2 term breaks the scaling property.

Q6. [Defining linearity · 09:00] What has been found when searching for nonlinear corrections to the Schrödinger equation?

A. Large corrections
B. Small but measurable corrections
C. None so far
D. Corrections only for light
Correct answer: C
Quantum mechanics appears to be exactly linear.

Q7. [Quantum mechanics as a framework · 00:00] How does the lecture describe quantum mechanics?

A. As a general framework that can be applied to different physical phenomena.
B. As a single experiment about light passing through two slits.
C. As a replacement for mathematics in physics.
D. As a theory that only applies to electromagnetism.
Correct answer: A
Quantum mechanics is a framework — a set of principles applied to many systems.

Q8. [Quantum mechanics as a framework · 00:30] What does quantum mechanics replace as the fundamental description of nature?

A. Thermodynamics
B. Classical physics
C. Relativity
D. Chemistry
Correct answer: B
It replaces classical physics; classical mechanics becomes an approximation.

Q9. [Quantum mechanics as a framework · 01:30] Applying quantum mechanics to electromagnetism gives…

A. General relativity
B. Statistical mechanics
C. Quantum electrodynamics
D. Fluid dynamics
Correct answer: C
QED is quantum mechanics applied to electromagnetism.

Q10. [Scaling and adding solutions · 11:50] If u1 and u2 solve a linear equation, which is also a solution?

A. u1 × u2
B. $u1^2$
C. a1·u1 + a2·u2
D. Only u1 − u2 when equal
Correct answer: C
Any linear combination of solutions is a solution.

Q11. [Scaling and adding solutions · 13:50] In quantum mechanics, the coefficients in a superposition can be…

A. Only positive integers
B. Complex numbers
C. Only zero or one
D. Only real fractions
Correct answer: B
Complex numbers are built into the framework.

Testing linear operators

Q12. [Addition · 00:30] How does a linear operator act on u + v?

A. L(u + v) = L(u) L(v)
B. L(u + v) = L(u) + L(v)
C. L(u + v) = L(u) − L(v)
D. L(u + v) = L(u)
Correct answer: B
Respecting addition means that the operator applied to a sum equals the sum of its outputs.

Q13. [Linear Combinations of Solutions · 01:30] If L(u) = 0 and L(v) = 0, what follows for a linear operator?

A. Only u is a solution.
B. Only v is a solution.
C. No linear combination is a solution.
D. Any linear combination of u and v is also a solution.
Correct answer: D
The transcript states that any linear combination is also a solution when both operator outputs are zero.

Q14. [Linear Combinations of Solutions · 01:30] Which condition is stated before concluding that any linear combination of u and v is also a solution?

A. L(u) and L(v) are both nonzero.
B. L(u) equals u and L(v) equals v.
C. L(u) and L(v) both equal zero.
D. L(u) and L(v) are unequal.
Correct answer: C
The stated condition is that both u and v have zero output under L.

Q15. [Maxwell Equations in Vacuum · 02:00] In vacuum, Maxwell equations are described as linear in which fields?

A. Electric fields only
B. Magnetic fields only
C. Neither electric nor magnetic fields
D. Both electric and magnetic fields
Correct answer: D
The transcript names both electric and magnetic fields.

Q16. [Maxwell Equations in Vacuum · 02:00] Which equations are identified as linear in electric and magnetic fields?

A. Maxwell equations in vacuum
B. Nonlinear equations generally
C. Only equations involving electric fields alone
D. Only equations involving magnetic fields alone
Correct answer: A
Maxwell equations in vacuum are the example given of equations linear in electric and magnetic fields.

Q17. [Nonlinear Equations · 02:30] What does the transcript say about nonlinear equations and superposition?

A. They always preserve superposition.
B. They do not generally preserve superposition.
C. They preserve superposition whenever magnetic fields appear.
D. They preserve superposition whenever electric fields appear.
Correct answer: B
The transcript says nonlinear equations do not generally preserve superposition, rather than making an absolute claim.

Q18. [Scaling · 00:00] Which identity expresses that a linear operator respects scaling?

A. L(a u) = a L(u)
B. L(a u) = L(u) + a
C. L(a u) = L(u)
D. L(a u) = a + u
Correct answer: A
The transcript states that scaling the input by a scales the output by a.

Q19. [Superposition · 01:00] Which two properties combine to give superposition?

A. Scaling and nonlinearity
B. Addition and nonlinearity
C. Scaling and addition
D. Zero outputs and nonlinearity
Correct answer: C
The transcript explicitly identifies scaling and addition as the properties that give superposition.